LearnerPath

End-to-End Technical Report

App flow · Architecture · Data pipeline · Retrieval logic · Cost

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1. What the App Does

LearnerPath is a personalized learning-path recommender. A learner picks subjects, chats briefly with an LLM to refine focus, takes an adaptive quiz that measures level per subject, and receives a study plan drawn from a curated database of ~33,000 courses across MIT, Harvard, Microsoft Learn, freeCodeCamp, YouTube, and Coursera. The path is customized to each learner's per-subject level, demonstrated skills, gaps, goal, budget, preferred formats, and time budget.

The four stages a user experiences:

- **Topics** — pick subjects, months, hours/day, goal, budget (free-only / free+paid), format & pace preferences.
- **Chat** — 2-4 turn LLM conversation refines focus_areas.
- **Quiz** — adaptive 2-round MCQ test, ~4 questions per subject per round. Round 2's difficulty per subject is targeted at each subject's Round 1 boundary.
- **Path** — per-subject level, per-subject strengths & gaps, week-by-week study plan, 4-8 recommended courses each tagged with concepts, level, format, price.

Key personalization principle

The retriever fires one query per subject at that subject's own level. So a learner who is **advanced in ML but beginner in Cybersecurity** receives an advanced ML course and a beginner Cybersecurity course — never a single blended level. Skills the learner already demonstrated in the quiz are not re-taught; skills they missed drive the retrieval boost.

2. System Architecture

2.1 Deployment topology

Component	Where deployed	Purpose	Cost/mo
Frontend (Vite + React + MSAL)	Azure Static Web Apps (Free)	SPA served to the browser	$0
Backend (FastAPI + uvicorn)	Azure App Service B1 Linux, Python 3.11	REST API for chat, assessment, score, session, plan	$13
Auth	Microsoft Entra External ID (LearnerPath tenant, ciamlogin.com)	OAuth2 / OIDC sign-in	$0
Database	Supabase Postgres, free tier (500 MB)	Unified **courses** table + **learning_sessions** table	$0
LLM	Azure OpenAI (learnerpathmodels resource)	gpt-4.1-mini for chat/quiz/planner/tagging; text-embedding-3-small for vectors	usage-based
CI/CD	GitHub Actions	azure-backend.yml + azure-static-web-apps-*.yml auto-deploy on push	$0

2.2 Request lifecycle for a full recommendation session

#	Step	Where
1	User visits SWA URL. MSAL redirects to Entra External ID.	Browser → Entra
2	After sign-in, browser calls **POST /api/session/start** with bearer token.	Browser → App Service
3	Topics form submitted → advances to Chat tab.	Browser only
4	**POST /api/chat** — Azure OpenAI (gpt-4.1-mini) with CHAT_SYSTEM prompt returns reply + focus_areas.	→ Azure OpenAI
5	When LLM sets ready_for_assessment=true, browser advances to Quiz.	Browser only
6	**POST /api/assessment** (round 1) — LLM generates ~4 MCQs per subject, each tagged with 1-3 concepts.	→ Azure OpenAI
7	**POST /api/assessment** (round 2) — adaptive: difficulty per subject targeted at Round 1 boundary.	→ Azure OpenAI
8	Quiz submitted → **POST /api/score**. This is the heaviest call.	→ Backend
9	Backend computes level_by_subject + missed_concepts_by_subject + demonstrated_concepts. Embeds one query per subject in parallel. Fires **match_courses** (keyword) + **search_courses_semantic** (HNSW vector) + local curated fallback in parallel per subject. Merges, deduplicates by URL and by concept-overlap. LLM planner picks 4-8 with weekly plan.	Backend → Supabase + Azure OpenAI

#	Step	Where
10	Backend saves the session to Supabase **learning_sessions**.	Backend → Supabase
11	Response returned to browser. UI renders study plan with price/level/format badges.	Browser

3. Database

3.1 Tables

Table	Rows	Purpose
courses	32,989	Unified catalog. All sources + concepts + embeddings live here.
learning_sessions	populated at runtime	One row per user session: topic_input, recommendation JSONB.

3.2 courses schema (after cleanup)

36 columns, grouped by purpose:

Column	Type	Purpose
id	uuid PK	internal
source	text	'harvard_pll' | 'ms_learn' | 'mit_learn' | 'freecodecamp' | 'youtube' | 'coursera'
external_id, url, canonical_url	text (UNIQUE source,url)	identity + cross-source dedupe key
title, description, provider, school, platform	text	display fields
level	text (beginner|intermediate|advanced)	filter (±1 tier)
topics, subjects, concepts, prerequisite_concepts, tags	text[]	GIN-indexed; drive retrieval matching
duration_hours, weeks, hours_per_week_min/max	numeric/int	time-budget filter (currently mostly null; ready for future use)
price_type	text (free|audit_free|paid|freemium)	budget filter
price_amount, price_currency, certificate_available, certificate_price	numeric/text/bool	financial info
format, pace, modality, language	text	delivery attributes
search_vector	tsvector	GIN full-text index
embedding	vector(1536)	HNSW-indexed for cosine similarity semantic search
active	bool	soft-delete flag
scraped_at, created_at, updated_at	timestamptz	housekeeping
image_url	text	UI thumbnail

Removed during cleanup (unused/always-null): year_published, year_updated, trailer_url, subtitles, estimated_difficulty, consensus_level, views_count, likes_count, enrollment_count, ratings_count, rating. Also dropped the legacy **harvard_pll_courses** table (data now in unified **courses**).

3.3 Indexes (18 total)

Index	Column(s)	Type
courses_pkey	id	btree (unique)
courses_source_url_key	source, url	btree (unique)
idx_courses_topics, _concepts, _subjects, _tags	arrays	GIN
idx_courses_search	search_vector	GIN
idx_courses_title_trgm	title (pg_trgm)	GIN
idx_courses_embedding	embedding	**HNSW** vector_cosine_ops, m=16, ef_construction=32
idx_courses_hot	(active, level, price_type) WHERE active	composite partial btree
idx_courses_source, _platform, _level, _price_type, _duration, _active, _canonical, _updated	single columns	btree

HNSW was chosen over ivfflat because Supabase free tier's `maintenance_work_mem` (32 MB) can't build ivfflat with lists=180 (~70 MB required). HNSW builds under 32 MB and gives ~5-10% better recall on this dataset size.

4. Data Pipeline — How 33k Courses Got In

4.1 Sources ingested

Source	Rows	Method	Auth
Coursera	23,575	Public catalog API (paginated JSON)	None
Microsoft Learn	4,595	learn.microsoft.com/api/catalog/ (single JSON dump)	None
MIT Learn (learn.mit.edu)	3,041	api.learn.mit.edu/v1/learning_resources_search/ (paginated)	None
YouTube (17 top edu channels)	1,159	YouTube Data API v3 — playlists.list	Free API key
Harvard PLL	521	JSON-LD scraper of pll.harvard.edu/catalog	None
freeCodeCamp	98	GitHub repo superblocks/*.json	None
Total in courses table	**32,989**		

Every ingestor lives at `scripts/sources/`. They share a common **base.py** helper (upsert on (source, url), exponential backoff, batch size 100) and are each idempotent + resumable. Re-running skips already-present rows.

4.2 Enrichment passes (LLM-added value)

Pass	Adds	Model	Cost	Status
Concept tagging	concepts[] + prerequisite_concepts[] per row	gpt-4.1-mini	~$4 total	■ 100% (32,989 rows)
Embeddings	vector(1536) per row	text-embedding-3-small	~$0.30 total	■ 100% (32,989 rows)

Concept tagging failed on ~209 rows (mostly YouTube playlists with empty descriptions or content-filter false-positives). Those rows carry sentinel concepts `_unparseable_` or `_filtered_` — they still surface via topic match, just not via concept match.

5. Retrieval Logic — How We Pick Courses

This is the core intellectual property. Every recommendation is built by combining keyword retrieval, semantic retrieval, curated fallback, and (only if nothing works) LLM-suggested web extras. The LLM is only a **ranker + explainer**, never a searcher — which is why URLs are always valid and latency is bounded.

5.1 Signals from the user (input to retrieval)

Signal	Source	Effect
subjects[]	TopicInput	Retrieval fires one parallel pass per subject
focus_areas[]	Chat	Included in topic overlap + semantic query text
level_by_subject	Quiz scoring	SQL filter: level IN (subject_level ±1 tier)
missed_concepts_by_subject	Quiz (wrong answers × MCQ.concepts)	Passed to match_courses as q_concepts (2× rank boost)
demonstrated_concepts	Quiz (correct answers × MCQ.concepts)	Fed to LLM planner: 'do not re-teach these'
budget	TopicInput (free_only | free_and_paid)	SQL filter: price_type IN (free, audit_free) if free-only
duration_months × hours_per_day	TopicInput	Passed as max_hours (currently light-touch; many rows have null duration)
language	Currently 'en' (single-language MVP)	SQL filter: language = 'en' OR language IS NULL
goal, preferred_formats, pace	TopicInput	Fed to planner prompt for downstream weighting

5.2 The three retrieval layers

Layer 1 — Keyword retrieval (Supabase RPC `match_courses`)

For each subject, fired in parallel with these params:

```
SELECT * FROM courses
WHERE active = true
AND level = ANY(nearby_levels(subject_level)) -- ±1 tier
AND price_type = ANY(allowed_by_budget)
AND language IN (learner_lang, 'en')
AND (topics && q_topics OR (q_concepts <> '{}' AND concepts && q_concepts))
ORDER BY
(2 if concepts overlap q_concepts else 0) +
(1 if topics overlap q_topics else 0) DESC,
updated_at DESC
LIMIT 15;
```

Ranking: concept overlap counts **double** a topic-only match. This is what makes gap-driven recommendation work — a learner who missed 'backpropagation' on the quiz gets courses tagged with that exact concept ranked above generic 'deep learning' overviews.

Layer 2 — Semantic retrieval (Supabase RPC `search_courses_semantic`)

In parallel with Layer 1, per subject, we embed a query text such as:

```
"Machine Learning. focus on neural networks. skills to learn: backpropagation, gradient descent"
```

This 1536-dim vector goes into a cosine-distance query over the HNSW index. Catches paraphrased or niche queries where exact keyword match misses (e.g. user gap 'chain rule for gradients' finds a course on 'backpropagation' even though the terms differ).

```
SELECT * FROM courses
WHERE active AND embedding IS NOT NULL
AND level = ANY(...) AND price_type = ANY(...)
ORDER BY embedding <=> q_embedding
LIMIT 15;
```

Layer 3 — Curated fallback (in-process Python)

`backend/app/catalog.py` holds ~60 hand-picked classic courses (MIT OCW playlists, Karpathy, 3B1B, Damodaran, Yale Sandel, etc.). Used as a safety net; matches by topic overlap only.

Layer 4 — LLM web fallback (only if pool < 12)

If keyword + semantic + curated return less than 12 candidates for a niche query, we ask the LLM to propose up to 3 well-known free resources **from a whitelisted domain set** (youtube.com, coursera.org, edx.org, mit.edu, stanford.edu, harvard.edu, cs50.harvard.edu, nptel.ac.in, khanacademy.org, freecodecamp.org, 3blue1brown.com, fast.ai). Every returned URL is validated with an HTTP HEAD request before being accepted. Hallucinated URLs never reach the user.

5.3 Dedup + merge

After all three layers return, we apply two dedup passes:

- **URL dedup** — same course from multiple layers/sources counted once.
- **Concept-overlap dedup** — drop later rows that share ≥60% concepts with an earlier one. This prevents 'Intro to Programming (4 wk)' and 'Programming Basics (3 wk)' both appearing.

Result: top 40 diverse, level-appropriate candidates handed to the planner.

6. The Planner — LLM as Ranker & Explainer

Feed the LLM: learner profile (subjects, level_by_subject, gaps, demonstrated_concepts, goal, budget, formats, pace, duration/hours), the 40 candidates with their concepts, and the RECOMMEND_SYSTEM prompt.

Prompt constraints

- Pick 4-8 from the candidates using **exact URLs**. Do not invent URLs.
- **No two courses may teach the same thing.** Every picked course must add material the others don't cover.
- Order foundation → intermediate → specialization → capstone/project.
- Every skill in the learner's GAPS list must be covered by at least one picked course; if no candidate covers a gap, add an extra from the whitelisted domains.
- Do NOT recommend material for concepts the learner already demonstrated.
- Respect budget: if free_only, silently drop paid candidates.
- Weight by goal: job → portfolio; certification → practice tests; project → hands-on; curiosity → conceptual survey; exam_prep → structured syllabi.
- Produce a **weekly_plan** array: one entry per week, each with focus, primary_resource, secondary_resource, checkpoint.
- Report strengths and gaps per subject.

Example output shape for a 6-month, 2 h/day path (ML intermediate, DL intermediate, Cybersec beginner):

```
Week 1-2: Cybersecurity foundations (Cybersec is beginner + all gaps)
Week 3-4: Cross-validation deep dive (ML gap)
Week 5-6: Deep learning fundamentals (DL gap: backprop)
Week 7-9: Transformers + attention (DL specialization)
Week 10-14: Applied ML capstone project (goal=job → portfolio)
Week 15-18: Cybersecurity intermediate
Week 19-24: End-to-end ML system w/ security review
```

7. Cost Breakdown

7.1 One-time enrichment (already spent)

Item	Cost
Ingesting 33k courses from 6 public APIs	$0 (no LLM used)
Concept tagging 32,989 rows @ gpt-4.1-mini	~$4
Embeddings 32,989 rows @ text-embedding-3-small	~$0.30
Runtime dev testing (chat, quiz, planner)	~$1-2
One-time total to date	**~$6**

7.2 Monthly recurring

Item	Monthly
Azure App Service B1 (backend)	$13
Azure Static Web Apps (frontend)	$0 (Free tier)
Supabase Postgres (DB + indexes + embeddings)	$0 (Free tier, ~40% of 500 MB used)
Microsoft Entra External ID (auth)	$0 (Free up to 50k MAU)
Azure OpenAI runtime @ 1,000 completed user sessions/mo	~$30-40 (gpt-4.1-mini planner + assessment + chat)
GitHub Actions CI/CD	$0 (public repo, 2k free min)
At 0 sessions/mo	**$13**
At 1,000 sessions/mo	**~$43-53**
At 10,000 sessions/mo	**~$310-410**

$200 Azure credit ÷ ~$45/mo = ~**4.5 months** of full production at 1,000 sessions/mo. Plenty of runway to validate and monetize.

7.3 Where the runtime money goes

- Each /api/score call embeds N subjects (~$0.00003) + retrieval SQL (~free) + LLM planner (~$0.02-0.03).
- Each /api/assessment call: ~$0.01 per round.
- Each /api/chat turn: ~$0.005.
- Per completed session end-to-end: **~$0.04-0.06**.

8. Status Board

Component	Status
Unified **courses** table + all indexes (incl. HNSW vector)	■ Done
Ingestion — 6 sources, 32,989 rows	■ Done
Concept tagging (32,989 rows, ~$4)	■ 100%
Embeddings (32,989 rows, ~$0.30)	■ 100%
Keyword retrieval RPC (match_courses)	■ Done
Semantic retrieval RPC (search_courses_semantic) + HNSW index	■ Done
Hybrid retriever: keyword + semantic + curated + LLM fallback, parallel per subject	■ Done
Adaptive 2-round quiz with per-subject stratification	■ Done
Concept-aware assessment (MCQs tagged with concepts)	■ Done
Per-subject level detection + level_by_subject → retrieval + planner	■ Done
Missed-concepts drive q_concepts boost in retrieval	■ Done
Dedup: URL + concept-overlap ≥60%	■ Done
Planner rules: no duplicate topics, foundation → capstone ordering, gaps must be covered	■ Done
Budget filter (free_only / free_and_paid)	■ Done
Frontend deployed to Azure Static Web Apps	■ Live
Backend deployed to Azure App Service	■ Live
MSAL / Entra External ID auth	■ Live
MVP quotas (single-login + IP + TTL)	■ Coded (currently disabled)
Repo cleanup (dead files, unused imports, unused columns)	■ Done (this branch)
User feedback loop (■/■ per course)	■ Planned
Prerequisite-safe filtering (require learner's demonstrated concepts ⊇ course prereqs)	■ Planned
Nightly ingest cron (GitHub Actions)	■ Planned

9. Repository Map (post-cleanup)

Path	Purpose
backend/app/main.py	FastAPI app: /api/chat, /api/assessment, /api/score, /api/session/start, /api/plan/{user_id}
backend/app/hybrid_retrieval.py	The orchestrator: parallel keyword + semantic + curated + LLM fallback.
backend/app/azure_client.py	OpenAI SDK wrapper (chat_json, chat_text) + embed_text for retrieval.
backend/app/prompts.py	CHAT_SYSTEM, ASSESSMENT_SYSTEM, RECOMMEND_SYSTEM.
backend/app/schemas.py	Pydantic types (TopicInput, MCQ with concepts[], Course, ...).
backend/app/catalog.py	~60 hand-picked classics used as fallback via filter_catalog.
backend/app/quota.py	MVP single-login / IP-block / 2-min TTL (currently disabled).
backend/app/auth.py	Entra JWT validation via JWKS.
backend/app/supabase_client.py	Supabase client + save_session / get_latest_recommendation.
backend/app/config.py	Pydantic-settings loader for backend/.env.
frontend/src/App.tsx	Root, 4-tab flow (Topics → Chat → Assessment → Results).
frontend/src/components/Tab*.tsx	Individual tab views.
frontend/src/lib/api.ts	Fetch wrapper with MSAL bearer token.
frontend/src/lib/authConfig.ts, useSessionQuota.ts, types.ts	MSAL config, session hook, TS types.
scripts/sources/base.py	Shared upsert / normalize helpers used by every ingestor.
scripts/sources/ingest_*.py	One file per source (harvard_pll deprecated, kept in git history).
scripts/enrich_concepts.py, enrich_embeddings.py	Async LLM enrichment passes.
scripts/run_tagging_until_done.sh, chain_tagging_then_embeddings.sh	Orchestration scripts.
scripts/supabase_unified_courses_schema.sql	Base schema + match_courses RPC.
scripts/supabase_post_ingest_indexes.sql	HNSW vector index + composite hot-path + semantic RPC.
scripts/supabase_cleanup_and_indexes.sql	Column pruning + rebuild RPCs (latest applied).
scripts/track_cost.py, generate_report_pdf.py	Cost tracker + this PDF generator.
.github/workflows/azure-backend.yml, azure-static-web-apps-*.yml	CI/CD to Azure.
backend/startup.sh, run.sh	Azure App Service startup + local dev launcher.
supabase/schema.sql	DDL for learning_sessions table.

Files deleted in cleanup

- harvard_pll_courses.json (leftover scraper dump)
- scripts/harvard_pll_catalog_entries.py, import_harvard_pll.py, import_harvard_pll_to_supabase.py
- scripts/scrape_harvard_pll.py, .harvard_pll_checkpoint.json, supabase_harvard_pll_schema.sql
- scripts/supabase_semantic_search.sql (merged into supabase_post_ingest_indexes.sql)
- backend/app/mit_learn.py (live MIT fetch — replaced by DB ingest)
- frontend/src/lib/supabase.ts (unused; auth is MSAL, DB access via backend)
- backend/app/__pycache__/, scripts/sources/__pycache__/, .DS_Store

Dead code trimmed

- hybrid_retrieval: removed unused CURATED import and dead SOURCES list.
- main.py: removed unused fetch_mit_courses import.
- courses table: dropped 11 always-null / unused columns.
- Legacy harvard_pll_courses table: dropped (data already in unified courses).

10. Why This Beats a Pure LLM Recommender

Property	Pure LLM (ChatGPT-style)	LearnerPath hybrid
URL validity	~60% (LLM hallucinates)	>99% (real DB rows + HEAD-validated fallback)
Latency per recommendation	8-15 s	3-5 s (LLM dominates; DB <200 ms)
Coverage	Whatever model remembers	32,989 real courses from 6 sources, extensible
Personalization	Prompt engineering only	Per-subject level + gaps + demonstrated skills + goal + budget + pace
Level correctness	Provider label only	Per-subject level + ±1 filter + concept intersection + gap-boost
Freshness	Frozen at model cutoff	Weekly re-ingest cron refreshes catalog from source APIs
Reproducibility	None (temperature-driven)	Deterministic ranking; LLM only re-orders top 40
Cost per 1k sessions	$80-150 (web search + long context)	$30-40

The key insight: LLMs are for composing and explaining. Retrieval and ranking belong in structured storage with explicit indexes. This app draws that line cleanly — the LLM never searches the web, never invents URLs, never gets to pick freely from millions of possible resources. It gets 40 pre-filtered, pre-ranked candidates and its job is to order and explain them into a coherent multi-week study plan.

— End of report —